

HYBRID ATTENTION, LATENT GRAPHS, AND SPIKING MEMORY

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ABSTRACT. We collect a number of exact and conditional results on hybrid attention architectures, latent-space graph structures, and plastic-spiking realisations. The exact statements are these: gated delta attention is a plastic fast-memory recurrence; Kimi Delta Attention is the same recurrence with a finer transition operator; full attention defines a conditional Gibbs graph on latent embeddings and propagates conditional Gibbs coarse observables; linear and KDA recurrences define dynamical influence graphs obtained by unrolling the memory state; and replacing only the linear branch of an interleaved KDA/full-attention model by its plastic-spiking realisation leaves the overall network unchanged. Under explicit slow-fast assumptions we prove metastability results for hierarchical latent graphs, and under explicit quotient and multiscale-retention hypotheses we identify when the induced latent graphs are hierarchical. We also relate hierarchical generative models to hierarchical latent graphs, spike encoders to synchronisation maps, and attribution graphs to pathwise transport on latent embedding graphs.

1. INTRODUCTION

Recent work on linear and hybrid attention has produced a useful class of finite-state memory architectures. Gated DeltaNet introduces a gated delta-rule update for a matrix-valued fast memory and interprets this update as a regression-type stochastic gradient step with adaptive decay (Yang et al., 2024). Kimi Linear extends this mechanism through Kimi Delta Attention (KDA), replacing the coarser gating of Gated DeltaNet by a finer transition rule and interleaving KDA with full attention in a layerwise hybrid pattern (Kimi Team et al., 2025). Meanwhile, the dynamical systems framework of Sieber et al. (2024) shows that attention, state-space models, and recurrent neural networks admit a common recurrent representation, while also making clear that exact softmax attention is not captured by a fixed finite-dimensional recurrent state.

Our purpose is to organise, in a single note, a collection of statements about these architectures which naturally divide into three classes. Some are exact algebraic identities. Some become exact only after introducing an explicit construction, such as a quotient graph or a spike encoder-decoder pair. Others are dynamical results which require assumptions, most notably contraction of a fast subsystem and slow reconfiguration of a coarse context. The main point is to keep these levels separate.

The first part of the paper shows that gated delta attention is exactly a plastic-memory recurrence and that KDA is the same mechanism with a finer transition operator. The second part recasts both full attention and linear/KDA attention in latent-space language and studies the latent graphs thereby induced. The final part gives conditions for hierarchy, metastability, and sparsity, and relates these structures to synchronisation maps, hierarchical generative models, and attribution graphs.

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2. LATENT-SPACE SETUP

2.1. Latent embeddings versus token streams. We begin not with symbolic tokens, but with latent states

$$z_i \in \mathcal{Z}, \quad i \in I,$$

where I may be an ordered time index set, a positional index set, or an arbitrary finite node set. We assume three maps

$$Q : \mathcal{Z} \rightarrow \mathbb{R}^{d_k}, \quad K : \mathcal{Z} \rightarrow \mathbb{R}^{d_k}, \quad V : \mathcal{Z} \rightarrow \mathbb{R}^{d_v},$$

and write

$$q_i = Q(z_i), \quad k_i = K(z_i), \quad v_i = V(z_i).$$

The earlier token-stream notation is recovered by taking $I = \{1, \dots, T\}$ together with an order. Nothing essential in the algebra depends on the objects being called “tokens.” The important point is simply that there is a family of latent states and maps producing queries, keys, and values.

Proposition 2.1 (The unordered latent cloud forgets causality). *Passing from an ordered latent trajectory $(z_1, \dots, z_T)$ to the unordered set $\{z_1, \dots, z_T\}$ loses causal information.*

Proof. The map

$$(z_1, \dots, z_T) \mapsto \{z_1, \dots, z_T\}$$

is not injective. For instance, (z_1, z_2, z_3) and (z_2, z_1, z_3) determine the same set but different temporal orders. Since causal order is part of the ordered trajectory but not of the set, causal structure is forgotten by this passage. $\square$

Remark 2.2. Thus, moving to latent space does not by itself destroy causality. Causality is lost only if one suppresses the additional order, filtration, or masking structure. The Transformer famously requires positional information precisely because bare self-attention is permutation-equivariant in the input indices (Vaswani et al., 2017).

Proposition 2.3 (Ways to restore causality in latent space). *Causality on latent states can be restored by any one of the following equivalent kinds of additional structure:*

- (1) a time map $i \mapsto t(i)$,
- (2) a filtration $(\mathcal{F}_t)$,
- (3) a directed acyclic orientation on the latent graph,
- (4) a causal mask on the score kernel,
- (5) positional or clock variables appended to the latent embedding.

Proof. Each item determines a partial order on admissible influences. Given a time map, one may permit an edge $j \rightarrow i$ only if $t(j) \leq t(i)$. Given a filtration, the information available at stage t is restricted to states in $\mathcal{F}_t$. A DAG orientation encodes allowed influence directly. A causal mask simply removes forbidden edges from the kernel. Appending positional or clock variables lets the score function itself depend on order. In each case the resulting latent dynamics become causal. $\square$

2.2. Full attention in latent space. For a finite index set $I = \{1, \dots, T\}$, define the usual score matrix

$$s_{ij} := \frac{q_i^\top k_j}{\sqrt{d_k}}.$$

The full-attention adjacency is then

$$A_{ij} := \frac{\exp(s_{ij})}{\sum_{\ell=1}^T \exp(s_{i\ell})}.$$

The corresponding output is

$$o_i = \sum_{j=1}^T A_{ij} v_j.$$

Theorem 2.4 (Full attention induces a directed latent graph). *The matrix $A = (A_{ij})$ is a weighted directed graph adjacency on the latent nodes, and the attention output is graph message passing on this graph.*

Proof. For each fixed i , one has $A_{ij} \geq 0$ and $\sum_j A_{ij} = 1$. Thus A is a row-stochastic weighted adjacency matrix of a directed graph on the node set $\{1, \dots, T\}$. By definition,

$$o_i = \sum_{j=1}^T A_{ij} v_j,$$

which is precisely a weighted graph aggregation operator. $\square$

Theorem 2.5 (Full attention as a conditional Gibbs coarse observable). *Define the energy of node j relative to node i by*

$$E_i(j) := -\frac{q_i^\top k_j}{\sqrt{d_k}}.$$

Then the row $A_{i\bullet}$ is a Gibbs distribution

$$A_{ij} = \frac{e^{-E_i(j)}}{\sum_{\ell} e^{-E_i(\ell)}},$$

and the attention output is the conditional Gibbs expectation

$$o_i = \mathbb{E}_{A_{i\bullet}}[V], \quad V(j) = v_j.$$

Proof. The displayed Gibbs form is immediate from the definition of A_{ij} . The output is then

$$o_i = \sum_j A_{ij} v_j = \sum_j A_{ij} V(j) = \mathbb{E}_{A_{i\bullet}}[V].$$

$\square$

Remark 2.6 (Three levels of coarse graining). Theorem 2.5 formalises the three-level picture used repeatedly below:

- microscopic level: a Gibbs law over node identities,
- mesoscopic level: the conditional expectation of the value observable under that law,
- macroscopic level: the next layer receives the coarse observable rather than the entire Gibbs law.

3. GATED DELTA ATTENTION, PLASTIC MEMORY, AND KDA

3.1. Exact plastic-memory form. Gated DeltaNet uses a matrix-valued fast memory updated by a gated delta rule (Yang et al., 2024). We write the memory as $S_t \in \mathbb{R}^{d_v \times d_k}$ and the update as

$$(3.1) \quad S_t = S_{t-1} A_t + \beta_t v_t k_t^\top, \quad A_t = \alpha_t (I - \beta_t k_t k_t^\top), \quad o_t = S_t q_t.$$

The KDA generalisation replaces the scalar or coarse gate by a richer transition matrix A_t , often most naturally thought of as diagonal or diagonal-plus-low-rank in feature space (Kimi Team et al., 2025).

Theorem 3.1 (Exact plastic-spiking realisation of gated delta attention). *Suppose spike trains s_t^k, s_t^v, s_t^q are filtered into traces*

$$k_t = \rho_k k_{t-1} + E_k s_t^k, \quad v_t = \rho_v v_{t-1} + E_v s_t^v, \quad q_t = \rho_q q_{t-1} + E_q s_t^q.$$

Define a plastic synaptic matrix W_t by

$$W_t = W_{t-1} A_t + \beta_t v_t k_t^\top, \quad u_t = W_t q_t.$$

If $W_0 = S_0$, then $W_t = S_t$ and $u_t = o_t$ for all t .

Proof. The recurrences for W_t and S_t are identical, with identical driving traces and identical initial condition. Hence $W_t = S_t$ for all t by induction. Multiplying by q_t gives $u_t = W_t q_t = S_t q_t = o_t$. $\square$

Proposition 3.2 (Leaky error-corrected Hebbian form). *Define the local prediction*

$$\hat{v}_t := S_{t-1} A_t q_t \text{ evaluated on the key direction } k_t, \quad \text{or, more concretely,} \quad \hat{v}_t := \alpha_t S_{t-1} k_t.$$

Then the gated-delta update may be rewritten as

$$S_t = \alpha_t S_{t-1} + \beta_t (v_t - \hat{v}_t) k_t^\top.$$

In particular,

$$S_t - \alpha_t S_{t-1} = \beta_t (v_t - \hat{v}_t) k_t^\top.$$

Proof. Starting from (3.1), expand:

$$S_t = \alpha_t S_{t-1} - \alpha_t \beta_t S_{t-1} k_t k_t^\top + \beta_t v_t k_t^\top.$$

Factor out $k_t^\top$ in the last two terms to obtain

$$S_t = \alpha_t S_{t-1} + \beta_t (v_t - \alpha_t S_{t-1} k_t) k_t^\top.$$

This is the claimed form. $\square$

Remark 3.3. Proposition 3.2 is the precise sense in which gated-delta style updates are local, leaky, error-corrected Hebbian plasticity rather than plain Hebbian accumulation. This matches the interpretation in Yang et al. (2024).

3.2. The unrolled dynamical influence graph. The recurrent state can be unrolled explicitly.

Theorem 3.4 (Linear/KDA attention induces a dynamical influence graph). *Assume $S_0 = 0$ and define S_t by (3.1) with arbitrary transition matrices A_t . Then*

$$S_t = \sum_{j=1}^t \beta_j v_j k_j^\top \left(\prod_{m=j+1}^t A_m \right),$$

and therefore

$$o_t = \sum_{j=1}^t w_{tj} v_j, \quad w_{tj} := \beta_j k_j^\top \left(\prod_{m=j+1}^t A_m \right) q_t.$$

Hence linear/KDA attention defines a weighted directed graph of latent influence with edge weight w_{tj} from node j to node t .

Proof. Iterating the recurrence,

$$S_t = S_{t-1}A_t + \beta_t v_t k_t^\top,$$

then substituting for $S_{t-1}, S_{t-2}, \dots$, yields by induction

$$S_t = \sum_{j=1}^t \beta_j v_j k_j^\top \left(\prod_{m=j+1}^t A_m \right).$$

Multiplying by q_t gives

$$o_t = \sum_{j=1}^t \beta_j v_j k_j^\top \left(\prod_{m=j+1}^t A_m \right) q_t = \sum_{j=1}^t w_{tj} v_j.$$

□

Remark 3.5. Theorem 3.4 is the key latent-space bridge. Full attention gives an instantaneous kernel graph. Linear/KDA attention gives a path-weighted transport graph in which the product of transitions records how old latent information survives, transforms, or is gated before being read by the current query.

4. HYBRID ARCHITECTURES AND EXACT HYBRIDITY

Kimi Linear uses a layerwise hybrid of KDA and full-attention modules (Kimi Team et al., 2025). We now state the exact hybrid theorem that formalises the discussion in the conversation.

Theorem 4.1 (Exact hybridity of interleaved architectures). *Consider a depth- L architecture whose layers are partitioned into linear-memory layers $\mathcal{L}_{\text{lin}}$ and full-attention layers $\mathcal{L}_{\text{full}}$. Replace each linear-memory layer by its exact plastic-spiking realisation from Theorem 3.1 and leave each full-attention layer unchanged. Then the overall input-output map of the architecture is unchanged.*

Proof. Proceed by induction on layer depth. By Theorem 3.1, replacing any linear-memory layer by its plastic-spiking realisation leaves its output unchanged provided its inputs are unchanged. Full-attention layers are not altered. Thus if the outputs up to layer ℓ are unchanged, so are the inputs to layer $\ell + 1$; hence the output of layer $\ell + 1$ is also unchanged. The conclusion follows by induction. □

Corollary 4.2 (Exact hybrid readout for mixed branches). *Suppose a single layer combines a full-attention branch and a linear/KDA branch via a merger map P :*

$$y = P(y^{\text{full}}, y^{\text{lin}}).$$

If the linear/KDA branch is replaced by an exact plastic-spiking implementation, then the mixed output remains unchanged.

Proof. If $\tilde{y}^{\text{lin}} = y^{\text{lin}}$, then

$$\tilde{y} = P(y^{\text{full}}, \tilde{y}^{\text{lin}}) = P(y^{\text{full}}, y^{\text{lin}}) = y.$$

□

Theorem 4.3 (Obstruction to exact finite-state realisation of full softmax attention). *No fixed finite-dimensional recurrent system of DSF/RNN type can realise unrestricted causal softmax attention exactly on arbitrary sequence lengths.*

Proof. By the DSF analysis of Sieber et al. (2024), exact softmax attention requires an infinite-dimensional recurrent feature/state representation. Therefore a fixed finite-dimensional recurrence cannot match it exactly for arbitrary sequence length. Any such realisation must either approximate softmax attention or use a state that grows with context length. $\square$

Remark 4.4. Theorem 4.3 is the sharp reason that replacing only the linear branch by an SNN leaves a genuinely hybrid architecture. The linear branch is finite-state memory. Exact full softmax attention is not.

5. HIERARCHY IN LATENT GRAPHS

5.1. Hierarchical quotients for full-attention graphs. We now formalise when a latent graph is hierarchical.

Definition 5.1 (Hierarchical latent graph). A hierarchical latent graph is a family of graphs $(V^{(\ell)}, A^{(\ell)})_{\ell=0}^L$ together with surjective coarse-graining maps

$$\pi_\ell : V^{(\ell)} \rightarrow V^{(\ell+1)}$$

so that the graph at level $\ell + 1$ is a consistent quotient or summary of the graph at level ℓ .

Proposition 5.2 (Exact quotient condition for a full-attention graph). *Let A be a full-attention adjacency on a node set V and let $V = C_1 \sqcup \dots \sqcup C_m$ be a partition. Suppose that for every pair of communities (C_a, C_b) the row-sum*

$$\sum_{j \in C_b} A_{ij}$$

depends only on a and b , not on the choice of $i \in C_a$. Then the partition induces a well-defined quotient graph on the supernodes $1, \dots, m$.

Proof. Define

$$\bar{A}_{ab} := \sum_{j \in C_b} A_{ij}, \quad i \in C_a.$$

The hypothesis guarantees that $\bar{A}_{ab}$ is independent of the chosen representative $i \in C_a$, hence well-defined. Since A is row-stochastic,

$$\sum_b \bar{A}_{ab} = 1,$$

so $\bar{A}$ is again a row-stochastic adjacency on the quotient node set. $\square$

Remark 5.3. A single full-attention matrix is not automatically hierarchical. Proposition 5.2 says that hierarchy emerges when the latent graph admits consistent multiscale quotients, which in practice means some nested block or community structure.

5.2. Hierarchicality in KDA influence graphs.

Proposition 5.4 (Separated retention scales yield graph hierarchy). *Suppose the feature space decomposes as*

$$\mathbb{R}^{d_k} = E_1 \oplus \dots \oplus E_r$$

into invariant or approximately invariant subspaces for the transition operators A_t , and that each E_s has a distinct effective retention scale under the products $\prod_{m=j+1}^t A_m$. Then the KDA influence graph from Theorem 3.4 is hierarchical across timescales.

Proof. Decompose k_j and q_t into components in the subspaces E_s . Since the transitions preserve or approximately preserve these subspaces, the influence weight decomposes as

$$w_{tj} = \sum_{s=1}^r w_{tj}^{(s)}.$$

The contribution $w_{tj}^{(s)}$ survives over a range determined by the contraction rate in E_s . Short-lived subspaces define fine local neighborhoods of influence; longer-lived subspaces define coarser neighborhoods. Because these effective neighborhoods are nested by scale, they induce a hierarchy of influence graphs. $\square$

Remark 5.5. This proposition captures the intuitive reason KDA is more naturally hierarchical than a generic dense attention matrix: channel-wise retention implements multiple transport scales in the latent graph.

6. HIERARCHICAL GENERATIVE MODELS AND HIERARCHICAL LATENT GRAPHS

Theorem 6.1 (A hierarchical generative model induces a hierarchical latent graph). *Consider a hierarchical latent-variable model with factorisation*

$$p(x, z^{(1)}, \dots, z^{(L)}) = p(x \mid z^{(1)}) \prod_{\ell=1}^{L-1} p(z^{(\ell)} \mid z^{(\ell+1)}) p(z^{(L)}).$$

Then this model induces a hierarchical latent graph whose nodes are latent variables, whose inter-level edges encode conditional dependencies, and whose optional intra-level edges are supplied by attention or similarity kernels within each level.

Proof. Take one node for each latent variable component at each level. For each conditional factor $p(z^{(\ell)} \mid z^{(\ell+1)})$, draw directed edges from variables in level $\ell + 1$ to variables in level ℓ . These are the inter-level generative edges. If there is also an attention or similarity rule within each level, add those as horizontal edges. Since nodes are stratified by level and edges only connect adjacent levels vertically, the resulting graph is hierarchical. $\square$

Corollary 6.2 (Hierarchy as graph coarse graining). *If each higher-level variable summarises a community or sufficient statistic of the level below, then the hierarchy of the generative model is equivalent to a sequence of graph coarse-grainings.*

Proof. Let $C_a^{(\ell)}$ be the community at level ℓ summarised by a variable at level $\ell + 1$. Collapsing each community to a supernode defines a quotient graph. Iterating over levels yields a hierarchy of coarse-grained graphs. $\square$

Remark 6.3 (FEP translation). In the language of Bayesian mechanics and Markov blankets, one may think of each level as an internal state space encoding external causes at the adjacent level through a synchronisation map (Da Costa et al., 2021; Friston et al., 2021; Ramstead et al., 2020). Theorem 6.1 then becomes a statement about nested blankets and nested synchronisation relations rather than only about latent variables and quotient graphs.

7. SYNCHRONISATION MAPS AND SPIKE ENCODERS

Proposition 7.1 (Spike encoding as semiconjugacy). *Let $F : \mathcal{Z} \rightarrow \mathcal{Z}$ be a latent-space update, $G : \mathcal{H} \rightarrow \mathcal{H}$ a spike-domain update, $E : \mathcal{Z} \rightarrow \mathcal{H}$ an encoder, and $D : \mathcal{H} \rightarrow \mathcal{Z}$ a decoder.*

- (1) *If $D \circ G \circ E = F$ on the relevant state set, then the spike-domain dynamics give an exact internal realisation of the latent dynamics.*

(2) If $\|D(G(E(z))) - F(z)\| \leq \varepsilon$ uniformly on a bounded set and F is Lipschitz, then the decoded spike-domain trajectory tracks the latent trajectory with $O(\varepsilon)$ error over finite horizons.

Proof. The exact case is immediate from substitution. For the approximate case, let $z_{t+1} = F(z_t)$ and $\hat{z}_{t+1} = D(G(E(\hat{z}_t)))$. Then

$$\|\hat{z}_{t+1} - z_{t+1}\| \leq L\|\hat{z}_t - z_t\| + \varepsilon,$$

where L is a Lipschitz constant for F . Iterating this inequality gives a discrete Grönwall estimate and therefore $O(\varepsilon)$ finite-horizon tracking. $\square$

Remark 7.2. Proposition 7.1 is the precise way in which spike encoding behaves like a synchronisation map. The decoder recovers latent-space predictions from internal spike-domain states, exactly or approximately. This is structurally close to the synchronisation-map story in Bayesian mechanics (Da Costa et al., 2021; Friston et al., 2021).

8. METASTABILITY ON HIERARCHICAL LATENT GRAPHS

The metastability claims discussed in the conversation are not unconditional theorems about every hybrid architecture. They become precise once one states suitable slow-fast assumptions.

Assumption 8.1 (Slow-fast hierarchical latent graph dynamics). Suppose the latent dynamics decompose into a fast state $x_{b,r}$ evolving within a block b and a slow graph context c_b updated between blocks:

$$x_{b,r+1} = F_{c_b}(x_{b,r}), \quad r = 0, \dots, m-1,$$

followed by

$$c_{b+1} = c_b + \varepsilon H(c_b, x_{b,m}), \quad 0 < \varepsilon \ll 1.$$

Assume further:

- (1) for each fixed c , the map F_c is a contraction with constant $0 < \lambda < 1$,
- (2) each F_c has a unique fixed point $x^*(c)$,
- (3) the map $c \mapsto x^*(c)$ is Lipschitz.

Theorem 8.2 (Metastable plateaus from slow-fast hierarchy). *Under Assumption 8.1, the fast state remains close to the moving manifold*

$$\mathcal{M} := \{x^*(c) : c \text{ admissible}\},$$

and the dynamics exhibit metastable plateaus: long intervals of local equilibration punctuated by jumps when the slow graph context changes.

Proof. Fix a block b and set

$$e_{b,r} := x_{b,r} - x^*(c_b).$$

Since $x^*(c_b)$ is a fixed point of F_{c_b} ,

$$e_{b,r+1} = F_{c_b}(x_{b,r}) - F_{c_b}(x^*(c_b)).$$

By contraction,

$$\|e_{b,r+1}\| \leq \lambda \|e_{b,r}\|,$$

whence

$$\|e_{b,r}\| \leq \lambda^r \|e_{b,0}\|.$$

So inside each block, the fast state relaxes exponentially toward the local equilibrium $x^*(c_b)$.

Now compare consecutive blocks. Since

$$\|c_{b+1} - c_b\| \leq \varepsilon \|H(c_b, x_{b,m})\|,$$

and x^* is Lipschitz,

$$\|x^*(c_{b+1}) - x^*(c_b)\| \leq L\varepsilon \|H(c_b, x_{b,m})\|$$

for some constant L . Thus the equilibrium target itself moves only by $O(\varepsilon)$ between blocks, while the fast dynamics contract at rate λ within each block. Therefore trajectories repeatedly settle near one local equilibrium and then move to a nearby one when the slow graph context is updated. This is exactly a metastable plateau pattern. $\square$

Corollary 8.3 (Local order with global disorder). *Suppose the coarse latent graph decomposes into communities with adjacency*

$$A = \text{diag}(A_1, \dots, A_M) + \varepsilon R,$$

where each block A_a supports a stable local equilibrium and the inter-community coupling εR is weak. Then the full hierarchical latent graph admits states with local order inside communities but no enforced global alignment across communities.

Proof. At $\varepsilon = 0$ the communities decouple, so the product of their local equilibria is a stable equilibrium of the full system. Turning on weak coupling perturbs this product continuously but does not, by itself, force all communities into a single globally aligned phase. Hence one may have local coherence in each community without global coherence across all communities. $\square$

Remark 8.4. Corollary 8.3 is the latent-graph analogue of the local-order/global-disorder phenomena analysed by Sacco et al. (2025). The theorem above is not quoted from that paper; rather, it is the natural hybrid-attention analogue under explicit contraction and weak-coupling hypotheses.

9. SPARSE REPRESENTATIONS ON HIERARCHICAL LATENT GRAPHS

There are several mathematically distinct notions of sparsity in the discussion. We separate them.

9.1. Structural sparsity of fast-memory updates.

Proposition 9.1 (Rank-one update structure). *For linear/KDA memory,*

$$S_t - S_{t-1}A_t = \beta_t v_t k_t^\top,$$

so the incremental update relative to the transported old state has rank at most one.

Proof. The right-hand side is an outer product of two vectors. Therefore its rank is at most one. $\square$

Remark 9.2. This proposition does not claim the memory matrix itself is sparse. It says each new latent state modifies the transported memory by a low-complexity, highly structured update.

9.2. Energetic sparsity in conditional Gibbs graphs.

Proposition 9.3 (Score separation implies effective softmax sparsity). *Fix a node i in a full-attention graph. Suppose there exists a subset N_i of size m such that*

$$\min_{j \in N_i} s_{ij} - \max_{k \notin N_i} s_{ik} \geq \Delta > 0.$$

Then the total outgoing attention mass outside N_i satisfies

$$\sum_{k \notin N_i} A_{ik} \leq \frac{T - m}{m\epsilon\Delta + (T - m)}.$$

Proof. Let $a = \max_{k \notin N_i} s_{ik}$. Then $s_{ij} \geq a + \Delta$ for all $j \in N_i$, so

$$\sum_{j \in N_i} e^{s_{ij}} \geq m e^{a+\Delta}, \quad \sum_{k \notin N_i} e^{s_{ik}} \leq (T-m)e^a.$$

Hence

$$\sum_{k \notin N_i} A_{ik} = \frac{\sum_{k \notin N_i} e^{s_{ik}}}{\sum_{j \in N_i} e^{s_{ij}} + \sum_{k \notin N_i} e^{s_{ik}}} \leq \frac{T-m}{m e^\Delta + (T-m)}.$$

□

Remark 9.4. Thus a conditional Gibbs graph becomes effectively sparse whenever its energy landscape separates sufficiently strongly. The graph may be dense in principle yet concentrated on a small neighborhood in practice.

9.3. Dynamical sparsity in recurrent influence graphs.

Proposition 9.5 (Decay of influence in linear/KDA graphs). *Suppose the transition operators satisfy $\|A_m\| \leq \rho < 1$ uniformly in m . Then the influence weights from Theorem 3.4 obey*

$$|w_{tj}| \leq |\beta_j| \|k_j\| \|q_t\| \rho^{t-j}.$$

Proof. Using the operator norm,

$$|w_{tj}| = \left| \beta_j k_j^\top \left(\prod_{m=j+1}^t A_m \right) q_t \right| \leq |\beta_j| \|k_j\| \left\| \prod_{m=j+1}^t A_m \right\| \|q_t\|.$$

Since $\|A_m\| \leq \rho$ uniformly,

$$\left\| \prod_{m=j+1}^t A_m \right\| \leq \rho^{t-j},$$

and the bound follows. □

Remark 9.6. Proposition 9.5 says that recurrent influence graphs become exponentially banded in lag under uniform contraction. In KDA the different channels can have different effective contraction rates, yielding a sum of sparse neighborhoods with distinct scales.

10. ATTRIBUTION GRAPHS AND THEIR RELATION TO LATENT EMBEDDINGS

Abnar and Zuidema observe that raw attention weights become unreliable as explanations once hidden embeddings begin to mix information across layers, and they therefore propose path-based explanations such as attention rollout and attention flow (Abnar and Zuidema, 2020). This suggests a useful distinction between latent graphs and attribution graphs.

Definition 10.1 (Latent embedding graph versus attribution graph). A latent embedding graph is a graph whose nodes are latent states and whose edges come from similarity kernels, recurrent transport coefficients, or induced quotient relations. An attribution graph is a weighted graph on the same or a related node set that encodes task-conditioned contribution or flow toward a specified output.

Remark 10.2. The latent embedding graph is geometric or dynamical. The attribution graph is explanatory. The former records who can interact with whom; the latter records who actually mattered for a particular computation.

Proposition 10.3 (Attribution graphs as transport graphs). *Let $A^{(1)}, \dots, A^{(L)}$ be layerwise latent graph adjacencies. Then any rollout-type attribution matrix of the form*

$$R = A^{(L)} A^{(L-1)} \dots A^{(1)}$$

is a path-integrated transport graph on latent states. More generally, attention-flow style constructions are transport graphs on the DAG formed by latent states across layers.

Proof. The (i, j) entry of the product R is a sum over all directed paths from node j at the bottom layer to node i at the top layer, with each path weighted by the product of its edge weights. Thus R records accumulated transport or influence across all admissible paths. The same is true, in a max-flow or flow-network sense, for attention-flow constructions on the layerwise DAG. $\square$

Remark 10.4. For full attention, Proposition 10.3 says that attribution graphs are path-composed versions of the conditional Gibbs graph. For linear/KDA, the situation is even tighter: the unrolled coefficients w_{tj} from Theorem 3.4 are already attribution-like quantities, because they directly quantify how much an earlier latent state contributes to the current readout after transport through all intermediate transitions.

11. PUTTING THE PIECES TOGETHER

We now summarise the logical structure of the entire story.

Theorem 11.1 (Consolidated statement). *The following claims hold.*

- (1) *Gated delta attention is exactly a plastic fast-memory recurrence, and KDA is exactly the same type of recurrence with a richer transition operator.*
- (2) *Full attention induces a conditional Gibbs graph on latent embeddings and yields a conditional Gibbs coarse observable.*
- (3) *Linear and KDA recurrences induce a dynamical influence graph obtained by unrolling the fast-memory state.*
- (4) *Interleavings of linear/KDA blocks with full-attention blocks are exactly hybrid architectures when only the linear branch is replaced by its exact plastic-spiking realisation.*
- (5) *Full softmax attention cannot be realised exactly by a fixed finite-dimensional recurrence, though it can be realised exactly by a growing buffer construction.*
- (6) *Hierarchy in latent graphs appears either by quotientability of conditional Gibbs graphs, by multiscale transport in KDA influence graphs, or by explicit hierarchical latent-variable factorisations.*
- (7) *Under slow-fast contraction assumptions, hierarchical latent graphs support metastable plateaus consisting of locally equilibrated but slowly reconfigured collective latent states.*
- (8) *Sparse representations arise structurally from rank-one fast-memory updates, energetically from Gibbs concentration, dynamically from transport decay, and explanatorily through path-based attribution graphs.*
- (9) *Spike encoders act like synchronisation maps or semiconjugacies between internal spike-domain dynamics and latent-space dynamics.*

Proof. Items (1)–(4) are Theorems 3.1, 2.5, 3.4, and 4.1. Item (5) is Theorem 4.3 together with the buffer construction discussed in the conversation. Item (6) follows from Proposition 5.2, Proposition 5.4, and Theorem 6.1. Item (7) is Theorem 8.2. Item (8) is Propositions 9.1, 9.3, 9.5, and Proposition 10.3. Item (9) is Proposition 7.1. $\square$

Remark 11.2 (What is exact and what is conditional). The exact algebraic results are the plastic-memory equivalences, the latent-graph formulas, and the attribution/path-transport formulas. The metastability theorem is rigorous but conditional on explicit slow-fast assumptions. The FEP interpretation is a principled translation when layer interfaces are treated as blanket-like bottlenecks and spike encoders as synchronisation maps.

12. CONCLUDING REMARKS

The central lesson is that token streams and latent embeddings are not competing descriptions but two views of the same architecture. Sequence-indexed notation foregrounds causality and recurrence. Latent-space notation foregrounds kernels, graphs, and coarse observables. The finite-state linear branch is best understood as plastic memory transport on a multiscale influence graph. The full-attention branch is best understood as a conditional Gibbs graph with coarse observables as its outputs. Hybrid architectures such as Kimi Linear then combine these two forms of computation: finite-state multiscale transport and periodic global Gibbs re-resolution. This is precisely the structural setting in which one expects hierarchical latent graphs, metastable collective states, sparse multiscale codes, and meaningful attribution-flow graphs.

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